Which of These Best Describes Multiple Choice Evaluation with LLMs? A) Forced B) Flawed C) Fixable D) All of the Above

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Abstract

Multiple choice question answering (MCQA) is popular for LLM evaluation due to its simplicity and human-like testing, but we argue for its reform. We first reveal flaws in MCQA's format, as it struggles to: 1) test generation/subjectivity; 2) match LLM use cases; and 3) fully test knowledge. We instead advocate for generative formats based on human testing—where LLMs construct and explain answers—better capturing user needs and knowledge while remaining easy to score. We then show even when MCQA is a useful format, its datasets suffer from: leakage; unanswerability; shortcuts; and saturation. In each issue, we give fixes from education, like rubrics to guide MCQ writing; scoring methods to bridle guessing; and Item Response Theory to build harder MCQs. Lastly, we discuss LLM errors in MCQA—robustness, biases, and unfaithful explanations—showing how our prior solutions better measure or address these issues. While we do not need to desert MCQA, we encourage more efforts in refining the task based on educational testing, advancing evaluations.

1 Questioning Multiple Choice Questions

Multiple choice question answering (MCQA) is the standard for large language model (LLM) evaluations, prized for simplicity and similarity to human testing (Robinson and Wingate, 2023). When designing new benchmarks, MCQA seems easy to implement (Guo et al., 2023), and when selecting new LLMs to use, MCQA leaderboards inform our decisions (Fourrier et al., 2024). If you want to build a popular dataset, prove your LLM is smart, or even publish a position paper, it is hard to avoid MCQA.

Standardized testing groups have long explored ways to better use MCQA for student testing (Angoff, 1971). But despite years of use in NLP (Turney et al., 2003), few have asked: 1) should MCQA be a standard model evaluation format; and 2) are its datasets well-designed? This position paper argues: **Evaluating LLMs with MCQA has flaws in both its inherent format and dataset construction.** We state our position in three points (Fig 1).

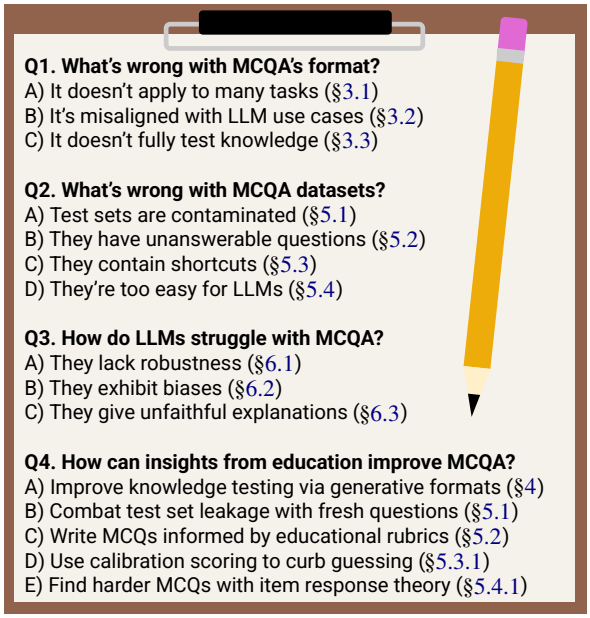


Figure 1: Overview of this paper. We show many problems in formats (§3), datasets (§5), and LLMs (§6) when using MCQA. Along the way, we propose solutions and ideas for future work, drawing from educational testing.

We first argue MCQA is not an ideal standardized format for LLM evaluations, showing its goal of "pick the best answer" cannot optimally test generation or subjectivity (§3.1), misaligns with LLM use cases (§3.2), and poorly tests knowledge (§3.3). Drawing from education, we advocate two *generative* improvements to MCQA's format for future exploration: 1) providing short, constructed-response answers without using choices (§4.1); and 2) evaluating explanations for model answers (§4.2). These formats capture generation or subjectivity, match LLM use cases, and improve knowledge testing, all while mostly preserving MCQA's simple scoring.

Next, we argue even when MCQA is a useful format, its datasets suffer from: dataset leakage (§5.1), unanswerable MCQs (§5.2), shortcuts (§5.3), and saturation (§5.4), degrading MCQA's utility. To

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enhance NLP dataset design, we offer solutions for each issue based on best practices in human testing, like rubrics to flag MCQ errors (§5.2), metrics to curb guessing from shortcuts (§5.3.1), and Item Response Theory (Baker, 2001) to cull shoddy MCQs and make the ones left more challenging (§5.4.1).

Lastly, we show many errors of LLMs in MCQA directly relate to MCQA's flaws (§6). These issues, like brittleness to perturbations (§6.1), bias toward certain options, cultures, and languages (§6.2), and generating unfaithful explanations (§6.3), can all be better measured or addressed with our proposed improvements to MCQA's format and datasets.

Many promising improvements to MCQA draw from education, a field dedicated to effective assessment (Haladyna et al., 2002), but these practices are rarely used in NLP. Adopting them demands more effort—MCQA is popular as it seems simple—but this effort is worth it to improve evaluations. To encourage researchers to take on these challenges, we conclude with guidelines for designing meaningful evaluations whether or not you use MCQA (§7).

2 Background: A Brief History of MCQA

A multiple-choice question (MCQ) is a question q and set of choices $\mathcal{C}$.[1] One choice $a \in \mathcal{C}$ is the gold answer, while others are plausible-sounding but incorrect distractors $\mathcal{D} = \mathcal{C} \setminus \{a\}$ meant to test misunderstandings.[2] MCQA's simple goal—picking the best answer a—is popular for LLM evaluation, but it has flaws. Before naming them, we first review its history in human testing (§2.1) and NLP (§2.2).

2.1 Why MCQA is the Standard for Humans

The MCQA format originated in 1914 with Frederick Kelley's Kansas Silent Reading Test (Kelly, 1916), proposed as an efficient measure of student reading comprehension (Monroe, 1917). Soon after, MCQA was attempted at scale, notably with Robert Yerkes' Army Alpha and Beta tests (Yerkes, 1918) in 1917 to assess U.S. Army intelligence. An initial bottleneck in MCQA was the manual effort needed for scoring, which researchers like Benjamin Wood and Reynold Johnson tackled by designing automated grading systems (Brennan and Clark, 1971; Wood and Johnson, 2001) with IBM.

Automatic scoring eventually enabled MCQA's popularity in primary/secondary education (Butler, 2018), college admissions (Daneman and Hannon, 2001), language proficiency (Jamieson et al., 2000), and even common tasks like driver's permit exams (Beanland et al., 2013) or employee training (Puhakainen and Siponen, 2010). Parallel to this, education researchers began exploring the best practices for writing high-quality MCQs (Morrison and Free, 2001; Campbell, 2011), authoring distractors (Pho et al., 2015; Gierl et al., 2017), and designing test settings (Rakes, 2008; Shute, 2013).

Despite MCQA's simplicity and popularity, organizations still critically assess its use in standardized testing. In the United States, the SAT removes unsound MCQ types,[3] and France's Baccalauréat uses long essay tasks over MCQA.[4] We argue LLM evaluation needs similar scrutiny and should draw from education to refine MCQA's format and data.

2.2 How MCQA Became Popular for LLMs

NLP first used MCQs from human exams; solving these with models that used external sources was considered part of an "AI grand challenge" (Reddy, 1988), as it required semantic (Turney et al., 2003; Veale, 2004) and factual understanding (Strickland, 2013; Clark et al., 2013). Other early MCQs from Winograd (Levesque et al., 2012) or COPA (Roemmele et al., 2011) tested commonsense reasoning over events and ambiguity in premises. Soon after, Richardson et al. (2013) designed MCTest for machine reading comprehension (MRC) via fiction text and MCQs. All tasks challenged models, but most MCQA work studied MRC (Lai et al., 2017).

With the advent of larger, neural LMs (Devlin et al., 2019), MCQA needed to become harder. Researchers expanded MRC to test numerical reasoning (Dua et al., 2019) and uncertainty (Rogers et al., 2020), and successfully scaled existing commonsense MCQs (Sakaguchi et al., 2021). New MCQs testing LM pre-training knowledge also grew popular, often using commonsense in daily tasks (Talmor et al., 2019; Bisk et al., 2020) and science exams (Mihaylov et al., 2018; Clark et al., 2018).

As LLMs improved in generation (Brown et al., 2020), MCQA evaluation changed; models usually scored MCQA choices independently, but Robinson and Wingate (2023) showed prompting LLMs

[1] Some choices can link to many other choices (e.g., "All of the above"), but these are discouraged (Haladyna et al., 2002).

[2] Extractive QA (e.g., SQuAD) and classification (e.g., NLI) can have a finite set of choices, but are fixed (i.e., labels) or have non-misleading distractors, so such tasks are not MCQA.

[3] https://blog.prepscholar.com/sat-analogies-and-comparisons-why-removed-what-replaced-them

[4] https://www.education.gouv.fr/reussir-au-lycee/le-baccalaureat-general-10457

Subjective Multiple-Choice Question

Question: Ash redeemed themselves after retaking the test they failed. How will Ash feel as a result?
Choices: (A) relieved (B) accomplished (C) proud 👍 👍 👍
Answer: (B) accomplished

Figure 2: Commonsense MCQ from Palta et al. (2024) where the choice rated most plausible by users is not the same as the gold answer; both are subjectively correct in varied contexts.

with the question and *all* choices was easy to score and matched human testing. It soon became standard to test LLMs with MCQs; companies used the task to parade their models (Achiam et al., 2023), some equating it to intelligence (Anthropic, 2024). This industry adoption incentivized researchers to write more MCQs across topics (Rein et al., 2024), languages, and modalities (Zhang et al., 2023).

Recent work critiques LLM evaluations generally, discussing reproducibility issues (Laskar et al., 2024), how it should be a distinct discipline (Chang et al., 2024), and its failure to predict deployment settings (Saxon et al., 2024). We similarly argue that while MCQA is simple and popular, the task has flaws in its format and datasets, many of which can be fixed using insights from education research.

3 MCQA is Flawed as a Standard Format

MCQA is a simple format for student testing, but educators find tradeoffs: it may not predict student success (Moneta-Koehler et al., 2017) or evaluate knowledge (Simkin and Kuechler, 2005). We argue that the same issues apply to NLP and thus, MCQA should not be considered a gold standard for LLM evaluation. Specifically, we discuss MCQA's rigid goal (§3.1), misalignment with real LLM use cases (§3.2), and limited testing of knowledge (§3.3).

3.1 "Pick the *Best* Answer" is Too Rigid

One of MCQA's key issues is its rigid goal: pick the best answer from a set of choices. While easy to score, both designs—the use of 1) one gold answer; and 2) input choices—limit MCQA's applicability.

First, one gold answer hinders MCQA's use for evaluating subjectivity (Finetti, 1965). Still, we use this format for commonsense (Bisk et al., 2020), morals (Yu et al., 2024b; Scherrer et al., 2023), and culture (§6.2), where many choices can be subjectively right (Figure 2). Palta et al. (2024) find users rate distractors in commonsense MCQs as the most plausible choice in over 20% of cases. Thus, extra care is needed to write MCQs for subjective tasks.

Second, picking from choices means MCQs test *validation*, useful in tasks like LLM-as-a-judge or re-ranking which must compare answers (Gu et al., 2024), but inhibiting tasks like writing and coding that require *generation* (Yu et al., 2022). One may argue MCQA proxies generation (if you pick good answers, you generate good ones), but LLMs lack validation/generation consistency (Li et al., 2024f; West et al., 2023; Balepur et al., 2025). Validation and generation are thus separate skills, so MCQA is a poor format for evaluating generation ability.

In all, MCQA best tests LLMs in objective validation, struggling with subjectivity and generation.

3.2 Users Rarely Ask LLMs to Solve MCQs

Many leaderboards aim to rank LLMs by their overall abilities, helping users select the best model for their needs (Xia et al., 2024). Hence, they should adopt tasks that mirror the popularity of user needs, giving higher ranks to models that can actually help users (Balepur et al., 2024c; Mozannar et al., 2025).

MCQA is over-represented versus how LLMs are used; 32% of the tasks in HELM (Perlitz et al., 2024), 71% in GPT-4's card (Achiam et al., 2023), and 79% in OpenLLM (Fourrier et al., 2024, Big Bench has 21 MCQA tasks) are MCQA. In contrast, Ouyang et al. (2023) find in ShareGPT's set of ChatGPT queries that nearly all user queries ask for free-form text; we estimate 7.2% are validation (4.3% evaluation and 2.9% comparison). Similarly, WildChat notes just 6.3% of their LLM queries are factual QA (Zhao et al., 2024).[5] Thus, over 90% of queries are likely generative tasks (code, writing, or explanations), which MCQs struggle to test (§3.1).

Informative evaluation suites must reflect LLM use cases. This is precisely why MCQA exams are waning in graduate admissions criteria: they cannot fully predict graduate school success (Sampson and Boyer, 2001). Similarly, over-representing MCQA in evaluations obscures which LLMs best aid users.

3.3 MCQA Does Not Fully Test Knowledge

While MCQA fails to match real user needs (§3.2), we hope the format tests basic skills for such needs, justifying its usage. MCQA is meant to test knowledge (Moss, 2001), and with input texts, comprehension (Farr et al., 1990), but work in education shows MCQA may be suboptimal for these goals.

MCQs mainly assess the basic knowledge levels in Bloom's Taxonomy of educational goals (Krathwohl, 2002): recalling, understanding, and applying knowledge (Simkin and Kuechler, 2005; Shin

[5] It is not explicitly stated if these are even validation tasks, so this is another *upper* bound of validation task prevalence.

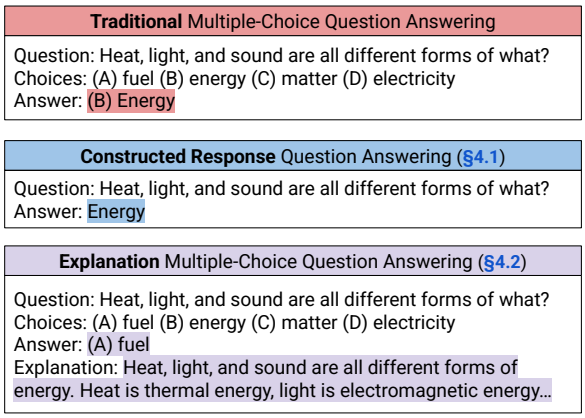


Figure 3: Example of adapting typical MCQs to our generative formats: Constructed Response and Justified MCQA.

et al., 2024); it is hard to write MCQs for the higher levels requiring reasoning (Stupans, 2006; Palmer and Devitt, 2007; Lin and Singh, 2012): analyzing, evaluating, and creating knowledge. As evidence, students can solve MCQs without full understanding, exposed in free-response answers (McKenna, 2019). MCQs with passages generally test comprehension, but some doubt this; Ozuru et al. (2013) find MCQA scores correlate with prior knowledge of the passage, overestimating true comprehension.

We believe these same insights can apply to NLP: MCQA may be apt for comprehension, but rewards LLMs for basic recall versus in-depth knowledge.

4 Generative MCQA Tasks are Promising

MCQA is flawed, so how should its role change? Standardized LLM evaluations must 1) proxy LLM use cases; and 2) test skills for (1). MCQA is unfit for (1), so evaluations need more tasks matching LLM needs (§3.2). Writing/explanation tasks are harder to score (Charney, 1984; Chakrabarty et al., 2022; Balepur et al., 2023a), but it is still odd they are often omitted, as they are typical needs (§3.2).

This limits MCQA to (2), but it is best for comprehension or validating objective facts (§3.1), not generation, subjectivity, or knowledge (§3.3). Validation/comprehension are valuable, which is why we discuss MCQA datasets in §5, but we need better formats for the other skills. Thus, we give two generative versions of MCQA to better test LLMs (Figure 3): Constructed Response (§4.1), answering sans choices, and Explanation MCQA (§4.2), justifying predictions. We ensure the formats only slightly increase evaluation complexity, mostly preserving MCQA's simplicity of scoring. We now describe the formats and future work to realize them.

4.1 Constructed Response Questions

Having LLMs solve MCQs without choices, called **Constructed Response (CR)** in education (Livingston, 2009) or short-form QA in NLP (Krishna et al., 2021), is one better format. CRQs test answer *generation* unlike MCQs (§3.1), better mirroring LLM needs (§3.2), and it is easier to write CRQs testing all skills in Bloom's Taxonomy (Krathwohl, 2002), so they better expose knowledge gaps (§3.3). Thus, students find CRQs harder than MCQs (Hancock, 1994), which can also delay saturation (§5.4).

Instead of writing CRQs to replace our vast existing MCQA data, a promising solution is to convert MCQs into CRQs by omitting choices and tasking LLMs to give a **short-form** answer $\hat{a}$ for question q, comparing it to gold answer $a \in \mathcal{C}$ (Bhakthavatsalam et al., 2021). Myrzakhan et al. (2024) show two hurdles in this: finding MCQs to convert[6] and scoring $\hat{a}$ with a. Recent efforts in flagging MCQ errors (Moore et al., 2023, 2024) and judging short-form answer correctness (Li et al., 2024g; Moore et al., 2022) showcase that we can realistically overcome these challenges. Thus, we believe combining this work with best practices for creating CRQs (Snow, 2012) can successfully implement the task.

4.2 Explanation Multiple Choice Questions

Constructed Response is promising (§4.1), but using one short answer is unfit for subjectivity (Lin et al., 2021) and conflicts user preferences for long outputs (Zheng et al., 2024b). We thus propose **Explanation MCQA (E-MCQA)** as another MCQA alternative from education (Lau et al., 2011): for a question q and choices $\mathcal{C}$, models give an answer $\hat{a} \in \mathcal{C}$ and explanation $\mathcal{E}$ for why $\hat{a}$ is right. This format tests generation (§3.1), matches the use case of explanations (§3.2), and has shown to test more knowledge levels over MCQA (Lee et al., 2011).

We envision E-MCQA being treated like reasoning tasks (Cobbe et al., 2021), checking if LLMs pick a like MCQA's simple scoring, but also studying $\mathcal{E}$. If models select a but justify it poorly, it exposes knowledge gaps like in student assessments (Jonassen and Kim, 2010), and when models give strong explanations for wrong answers, it enables partial credit for subjective tasks (Lau et al., 2011).

E-MCQA has many benefits, but needs metrics to score "good" explanations over many facets (Xu et al., 2023), like factuality to curb hallucinations (Min et al., 2023; Balepur et al., 2023b), plausibil-

[6] "Which of these best..." MCQs require using all choices.

ity for convincingness (Liu et al., 2023), and faithfulness to verify $\mathcal{E}$ supports a (Paul et al., 2024). We believe these goals could be achieved by merging ongoing efforts in building verifiers for LLM reasoning (Ling et al., 2023) with educational best practices for grading justifications (Jonassen and Kim, 2010), yielding reliable metrics that realize E-MCQA's potential. Kim et al. (2025) have made notable progress for this goal—building LLM judges to score justifications across various benchmarks—showing that implementing E-MCQA is feasible.

5 MCQA Datasets are Flawed but Fixable

MCQA is not always the best format (§3), but we still need high-quality MCQs for comprehension/-validation as well as tasks like LLM-as-a-judge (Gu et al., 2025) and re-ranking (Ma et al., 2023) which require comparing answers. Further, our generative MCQA formats (§4) still use MCQs as inputs.

However, like most NLP tasks, MCQA datasets have quality issues that impede their utility: leakage (§5.1), unanswerability (§5.2), shortcuts (§5.3), and saturation (§5.4). We now show how educators' solutions to these issues can inform NLP datasets.

5.1 LLMs Peek at MCQA Answer Keys

To build an MCQA dataset, we first need sources to write or collect MCQs. But as many sources end up being leaked[7] in LLM training data (Magar and Schwartz, 2022), such MCQs may confuse generalization abilities for memorization (Lewis et al., 2021). Private test sets (Sap et al., 2019) and decontamination (Zhou et al., 2023) help, but LLMs tuned on newer data can overlap with (1), and opacity in LLM data (Soldaini et al., 2024) blocks (2).

An ambitious solution to test set leakage is live MCQs that update over time to stay unseen (White et al., 2025), like how educators rewrite exams to impede cheating. Trivia (Jennings, 2007) and standardized testing groups frequently write new questions, making them ideal partners. To aid both parties, researchers could offer these groups tools for tutoring (Siyan et al., 2024), MCQ validation (Yu et al., 2024a), or answer scoring (Yang et al., 2020).

Test set leakage would be easier to fix if model designers released training data, but as we all know, most do not. While it is harder to design solutions for leakage that do not need training data, we hope researchers view it as a challenging, impactful research problem in evaluation and generalization.

[7] gpt-3 has seen 45% of RACE's test set (Sainz et al., 2023).

Unanswerable Multiple Choice Question

Question: The number of energy levels for the 55Mn nuclide are:
Choices: (A) 3 (B) 5 (C) 8 (D) 4
Answer: (A) 3

Rubric Errors

5. Use good grammar, punctuation, and spelling consistently ❌
13. Avoid over specific knowledge when developing the item ❌
24. Place options in logical or numerical order ❌
37. Make sure there is one and only one correct option ❌

Figure 4: Example unanswerable MCQ from MMLU (Gema et al., 2025), along with rubric criteria from Haladyna and Downing (1989) flagged by OpenAI's o1 (Jaech et al., 2024).

5.2 Some MCQs Have No Correct Answer

Once a source is found (§5.1), researchers collect or write MCQs, but errors often arise rendering them unanswerable, like mislabeling (Explained, 2023), multiple correct choices (Palta et al., 2024), ambiguity (Gema et al., 2025), missing contexts (Wang et al., 2024b), and grammar errors (Chen, 2023).

Educators write MCQs with rigorous protocols, and we must meet similar standards in NLP (Boyd-Graber and Börschinger, 2020); we should use educators' rubrics (Figure 4) for writing and validating MCQs (Haladyna and Downing, 1989). Such guidelines also specifically exist for distractors (Haladyna et al., 2002)—the part of MCQs that discern testees' skills—ensuring they are truly wrong, shortcut-proof (§5.3), and not too easy to rule out (§5.4). Beyond MCQ writing, rubrics can form data cards (Pushkarna et al., 2022) to help researchers record errors in their data and how they fixed them.

Recent work in LLM checklist evaluation (Cook et al., 2024), MCQ metrics (Moon et al., 2022), and MCQ generation (Feng et al., 2024) show parts of this workflow can be automated (Figure 4). Wang et al. (2024b) fix errors in MMLU by using LLMs to detect issues and write new choices. LLM judges can be inaccurate and biased (Xu et al., 2024b), so human-AI collaboration, like model-assisted refinement (Shankar et al., 2024) and task routing (Miranda et al., 2024), may be more promising. Errors will arise in MCQ writing, but educators' rubrics can help find and fix them, ensuring answerability.

5.3 MCQA Shortcuts May Let LLMs "Cheat"

Answerable MCQs (§5.2) are not always high quality, as shortcuts may let LLMs guess the answers to MCQs without knowing the answers (Du et al., 2023), overestimating model accuracy (Wiegreffe and Marasovic, 2021). Shortcuts arise from annotator artifacts (Gururangan et al., 2018), spurious patterns (Zhou et al., 2024b), or bypassed reason-

ing steps (Chen and Durrett, 2019). If LLMs best random guessing via partial inputs (Richardson and Sabharwal, 2020, e.g., choices only), they exist.

Below, we discuss how scoring (§5.3.1) and data collection methods (§5.3.2) can mitigate shortcuts.

5.3.1 Calibrated Scoring Can Deter Guessing

While rare in human testing,[8] scoring methods can penalize wrong guesses (Lau et al., 2011): **1) Probability scoring:** elicit confidence scores for each choice (Finetti, 1965); **2) Negative marking:** subtract points for wrong answers with abstention allowed (Holt, 2006); and **3) Elimination scoring:** students iteratively remove wrong choices until unsure (Ben-Simon et al., 1997). Confidence (Li et al., 2024c), abstention (Góral et al., 2024), and elimination (Ma and Du, 2023) have been studied in LLMs, so they may be easy to use for MCQA evaluation.

Calibration methods deter guessing, but also reward models that know their knowledge gaps (Guo et al., 2017). Such scoring methods are often ignored in evaluation (Bommasani et al., 2023), but it could let MCQA better test decision-making (Liu et al., 2025) and enable partial credit for subjective tasks where many choices may be tenable (§3.3).

5.3.2 We Can Design Shortcut-Proof MCQs

Data designers should limit shortcuts; an easy way is uniform design. When solving MCQs with only choices, Balepur et al. (2024b) find LLMs may exploit distributional differences, so like educators do (§5.2), we should write parts of MCQs consistently: via the same agent, source, and decoding method. HellaSwag (Zellers et al., 2019) leads to the highest known choices-only accuracy, where *user*-written answers and *model*-written distractors are inconsistent, showing the necessity of uniform data design.

Contrast sets are another tool that detects if models ignore inputs and use shortcuts (Gardner et al., 2020). In MCQA, they are entry pairs differing by some inputs (e.g., question) that change the answer (Figure 5), ensuring models attend to the perturbed input (Elazar et al., 2024). Balepur and Rudinger (2024) use contrast sets in commonsense MCQA to ensure none of their LLMs rely on shortcuts in choices to rank highly. Contrast sets are often made manually (Kaushik et al., 2020), so future work can test automatic ways to build them (Li et al., 2020).

Studying how users and models *"cheat"* (Saxon et al., 2023) in MCQs also finds shortcuts. In reading comprehension, Pang et al. (2021) give users `ctrl+F` to detect MCQs quickly solvable without using the full text, while Malaviya et al. (2022) flag MCQs where users can use simple heuristics. We can also train models to cheat; adversarial filtering trains simple models (e.g., bag-of-words) and omits MCQs they can solve (Zellers et al., 2018). Extending this, we believe having strong LLMs reason to cheat—similar to safety work in alignment faking (Greenblatt et al., 2024)—can help find shortcuts.

MCQ Paired Example A

Question: Some aerosols can decrease temperatures by blocking what?
Choices: (A) the sun (B) rain
Answer: (A) the sun

MCQ Paired Example B

Question: Which of the following increases moisture?
Choices: (A) the sun (B) rain
Answer: (B) rain

Figure 5: Example MCQ pair for a contrast set from Balepur and Rudinger (2024). The choices are identical, but the question swaps the answer, testing if models ignore the question.

5.4 LLMs Inevitably Ace MCQA Datasets

Even if we fix all of these issues, MCQs become too easy over time (i.e., saturated), no longer tracking LLM progress (Li et al., 2024d). To still use "easy" MCQA datasets, we need to make them harder for models. Below, we show how understanding which MCQs are hard (§5.4.1) and helping users author hard, interpretable MCQs (§5.4.2) delay saturation.

5.4.1 IRT Reveals Challenging MCQs

When LLMs excel in MCQA datasets, some MCQs remain hard; finding them and why they are hard informs data design (Sugawara et al., 2018, 2022). One MCQ difficulty metric is success rate (SR)—the number of models answering correctly (Gupta et al., 2025b)—but SR omits *which* models succeed. MCQs solved by just the worst or best model have equally low success rates, but the former suggests MCQ errors (§5.2)—as a weaker model besting all others is rare—while the latter matches our expectation. As SR conflates these cases, it cannot separate flawed MCQs from those discerning model ability.

Item Response Theory (IRT)—a tool used in education (Lord and Novick, 2008)—is a more robust way to find hard MCQs. While SR treats all models equally, IRT learns the skill of models to then estimate every MCQ's difficulty (how hard it is) and discriminability (how well it discerns between weak/strong models). IRT can then be used to fil-

[8] Since they may induce stress or anxiety in students (Vanderoost et al., 2018), but we think it is fine to stress LLMs.

Obscure MCQ

Q: In Spongebob, Tony's house has a poster referencing what band?
(A) Queen (B) Gorillaz (C) Kiss
Answer: (A) Queen

Adversarial MCQ

Q: How many non-pet characters live in SpongeBob's neighborhood?
(A) 3 (B) 4 (C) 5
Answer: (B) 4

Figure 6: Obscure and adversarial MCQs for *Spongebob Squarepants* inspired by Sung et al. (2025). GPT-4o answers both wrong (answer in blue). The former tests niche knowledge, but the latter is easy for those who have seen the show.

ter high-difficulty, high-discriminability MCQs as harder data splits (Polo et al., 2024), flag saturation if all MCQs have low difficulty/discriminability (Vania et al., 2021), and omit faulty MCQs with negative discriminability (Rodriguez et al., 2021).

While IRT-based filtering finds harder MCQs, it does not give new MCQs, limiting its long-term use. However, we can extend IRT to multi-dimensional IRT (Reckase, 2006, MIRT) to capture *many* latent skills, offering more insights into model abilities; by interpreting these skill dimensions, we can pinpoint model issues that inform future data efforts (§5.4.2). Gor et al. (2024) reveal LLM errors in abductive reasoning via a variant of MIRT—an issue confirmed by abduction research (Del and Fishel, 2023; Nguyen et al., 2023). MIRT could similarly find difficult MCQA topics, distractor patterns, or reasoning types for models (Benedetto et al., 2021).

Overall, IRT can find which MCQs are hard but also why, informing future data collection efforts.

5.4.2 A Good MCQ is Hard *and* Interpretable

A popular way to write harder MCQs is requiring obscure knowledge (Figure 6, left), sourcing from experts (Rein et al., 2024; Phan et al., 2025) and global competitions (Fang et al., 2024). These challenge models *and* humans, which is useful for AI safety work in scalable oversight (Bowman et al., 2022). However, this makes them uninterpretable for non-experts diagnosing model errors, especially when studying model rationales (§6.3). If LLMs err on obscure MCQs, it is hard for non-experts to find if errors are from faulty reasoning, misunderstandings, or knowledge gaps (Anderson et al., 2025).

While authors know which MCQs elude humans, writing ones that surface model errors while staying human-interpretable needs support. This is **adversarial** data collection's goal (Kiela et al., 2021)—building UIs to help authors write examples hard for models but easy for humans (Figure 6, right). Rather than using niche knowledge, authors must write MCQs with spurious patterns, misleading distractors (§5.2), or reasoning traps that trick LLMs but not humans (Xu et al., 2024c). As a result, these MCQs better expose robustness (§6.1) and logical reasoning (§6.3) errors, less clouded by knowledge.

Original Multiple-Choice Question

Q: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Kiss (B) Gorillaz (C) Queen
Answer: (A) Kiss

Robustness (Symbol Changing)

Q: In Spongebob, Tony's house has a poster referencing what band?
Choices: (P) Kiss (Q) Gorillaz (R) Queen
Answer: (R) Queen

Biases (Positions / Symbols)

Q: In Spongebob, Tony's house...
(A) Gorillaz (B) Queen (C) Kiss
Answer: (A) Gorillaz

Q: In Spongebob, Tony's house...
(A) Queen (B) Gorillaz (C) Kiss
Answer: (A) Queen

Explanations (Hallucinations)

Q: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Kiss (B) Gorillaz (C) Queen
Answer: ...If you meant Tony Fast Jr. (a minor character from the episode *Tutor Sauce*), his room has a poster that appears to reference the band Kiss. Thus, the answer is: (A) Kiss

Figure 7: LLMs err with robustness (e.g. inconsistent after shuffling), biases (e.g. favor symbols), and explanations (e.g. fail to justify answers) in MCQA. Correct answers are blue.

Adversarial MCQs are useful, but finding users to write high-quality ones is tough. Gamification—making the task fun—helps, used in building adversarial commonsense (Talmor et al., 2021), QA (Wallace et al., 2019; Sung et al., 2025), and fact-checking (Wallace et al., 2019) data. Wallace et al. (2022) use gamification to get adversarial NLI data and show it has *long-term* difficulty, delaying saturation. For more engagement, researchers can stir users to author MCQs exposing shocking failures, inspired by jailbreaking, where provocative outputs are naturally fun to elicit (Schulhoff et al., 2023).

Obscure and adversarial MCQs can both make MCQA harder: the former tests niche knowledge, while the latter better find reasoning or consistency failures that are uneclipsed by knowledge gaps.

6 Fixing MCQA Can Help Us Fix LLMs

Fixing issues in MCQA's format (§3) and datasets (§5) will not just improve evaluation quality; they can also improve our understanding of LLM weaknesses. This section outlines three persistent issues of LLMs in MCQA (Figure 7)—robustness (§6.1), bias (§6.2), and explanations (§6.3)—and how our prior solutions can better address or evaluate them.

6.1 New Prompts Lead to New MCQA Scores

On specific prompts, LLMs score highly in MCQA, but now crumble after prompts change, sensitive to:

choice symbols (Alzahrani et al., 2024), choice ordering (Zheng et al., 2024a), and phrasing (Wiegreffe et al., 2023). This brittleness degrades MCQA leaderboard reproducibility, as different evaluation setups yield conflicting rankings (Gu et al., 2025)

LLM robustness varies by task setup. In MCQA, early LLMs with poor instruction following (Zhang et al., 2022) used probability-based *scoring*, while instruction-tuning enabled LLMs to *generate* answers (Longpre et al., 2023); while logically equivalent, these give varying answers (Lyu et al., 2024). Probability scoring seems more at fault—more sensitive to prompts (Wang et al., 2024a)—doubting if LLMs can aid decision-making tasks that need accurate confidence scores (Liu et al., 2025). To track this progress in LLMs, researchers can use MCQA scoring protocols that measure calibration (§5.3.1).

As accuracy often *drops* post-perturbation (Zhou et al., 2024a), these errors in generalization could indicate dataset leakage (§5.1) or over-reliance on biases (discussed in §6.2). Another proposed explanation is symbol binding error: LLMs "know" the answer but cannot link it to the right choice (Wiegreffe et al., 2024; Xue et al., 2024). These failures weaken MCQA's ability to evaluate knowledge, obscured by memorization and symbol binding.[9] Our proposed solutions—like curating live MCQs for data leakage (§5.1) and generative MCQA formats to expose knowledge gaps without needing symbol binding (§4)—can more reliably assess knowledge.

6.2 LLMs are Biased MCQA Test-Takers

Like most NLP tasks (Baumler and Rudinger, 2022; Chu et al., 2024), LLMs show biases in MCQA, grouped into two main types. The first are MCQA-specific biases—selecting answers based on symbols (Zheng et al., 2024a), positions (Li et al., 2024e; Wei et al., 2024), or phrases such as "none of the above" (Xu et al., 2024a; Wang et al., 2025) rather than MCQ content. This degrades robustness (§6.1), masking model knowledge. We believe they likely stem from shortcuts (§5.3); LLMs tuned on data where choices with patterns are often correct will manifest these biases (Pacchiardi et al., 2024). Our safeguards—uniform design, contrast sets, and "cheating" (§5.3.2)—can reduce these biases, and more work in shortcuts may be key to quash them.

The second bias type stems from general training, like cultural/linguistic bias (Myung et al., 2024; Li et al., 2024a); LLMs often err on non-Western cultural MCQs (Acquaye et al., 2024; Azime et al., 2025) and non-English MCQs (Son et al., 2025; Li et al., 2024b). MCQs are popular for assessing bias (Guo et al., 2023), but if they use subjective commonsense (Seo et al., 2024), they are subpar (§3.1). Thus, we advise writing MCQs with rubrics to limit ambiguity and correct distractors (§5.2), ensuring bias is more objectively tested. If subjectivity persists, evaluators should consider strategies to manage it like Explanation MCQA (§4.2) or calibration scoring (§5.3.1), aiding bias testing and for E-MCQA, better matching how bias presents downstream (Seshadri and Goldfarb-Tarrant, 2025).

Lastly, to construct non-English MCQs, we discover researchers typically either: 1) collect MCQs in the desired target language (Son et al., 2025); or 2) translate English ones (Achiam et al., 2023); However, (1) may be infeasible for languages that are low-resource or where MCQA is rarely used,[10] while (2) may introduce error propagation during translation (Singh et al., 2024)—showing neither method is perfect on its own. Thus, researchers can adopt our prior solutions for both methods—like using rubrics during translation (§5.2) or collecting non-English MCQs from unseen sources (§5.1)—improving evaluations of multilingual abilities.

6.3 LLMs Struggle to "Explain Their Work"

Even if LLMs get the right answer to an MCQ, they may justify their selection **unfaithfully**—failing to mirror their true reasoning—whether via chain-of-thought (Lyu et al., 2023; Turpin et al., 2024) or self-explanations (Kim et al., 2024; Madsen et al., 2024). However, they are convincing, besting crowdworkers in judged quality (Mishra et al., 2024) and misleading users when wrong (Si et al., 2024), likely as standard alignment strategies tend to optimize LLMs for user preferences over helpfulness (Balepur et al., 2024c; Wen et al., 2025).

LLM explanation flaws are even clearer after logical consistency checks. Kawabata and Sugawara (2023) show LLMs often inaccurately explain answers to subquestions in reading comprehension, even when answering the higher-level question correctly. Similarly, Balepur et al. (2024a) find LLMs struggle to reason why distractors are wrong. These issues likely stem from broader LLM logical inconsistencies (Liu et al., 2024; Varshney et al., 2024).

[9] Such evaluations have downstream use (e.g., privacy, interpretability), but do not measure knowledge as intended.

[10] In Germany it is uncommon to test students with MCQs: https://www.reddit.com/r/AskAnAmerican/comments/rjxns4/

Explanations are popular LLM use cases (§3.2), highlighting the need for improved MCQA formats such as Explanation MCQA (§4.2) which explicitly test explanation skills. Further, LLMs' logically inconsistent explanations offer a path toward harder datasets (§5.4); tools like MIRT (§5.4.1) can identify logical error types (negation, decomposition) that elude LLMs, while adversarial collection can curate MCQs to excise these errors while staying easy for humans (§5.4.2). In all, LLMs' poor explanation abilities give an opportunity to design harder MCQA evaluations better aligned with user needs.

7 Call to Action: Benchmarking 101

If you want to make the best benchmark ever, where do you begin? First, define the ability you want to test and decide if MCQA is the right format (§3). If the ability matches a downstream task (e.g., coding), just use that task (Saxon et al., 2024). If the ability is fundamental (e.g., knowledge), consult education research to weigh alternative formats (§4).

If MCQA is the best format, find a data source to curb leakage—one with fresh content (§5.1). When curating MCQs from your source, follow educators' rubrics to ensure answerability (§5.2), and release the rubric as a data card to record errors (Pushkarna et al., 2022). Consistent design choices will limit shortcuts (§5.3), and providing a contrast set could help researchers check if their models over-rely on shortcuts (Gardner et al., 2020). As another safeguard, your benchmark can use calibration scoring beyond accuracy to discourage guessing (§5.3.1).

Post-release, models will hill-climb and saturate your data over time (§5.4). If you want to delay saturation, you may restart with an obscure knowledge source, but if you want your data to better diagnose errors, aim for interpretability. Use IRT (§5.4.1) to find which of your MCQs are hard and why, then design an engaging, adversarial dataset collection protocol (§5.4.2) guided by these insights, yielding a new dataset hard for models but easy for humans.

By using even some of educators' insights, we can refine the utility of MCQA—or any task. This approach takes more effort than the simple MCQA practices that initially attracted researchers, but if we do not address the flaws of MCQA, **what model abilities can our MCQA benchmarks even test?**

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8 Limitations

Inspired by Saxon et al. (2024), we organize our limitations section as potential counterarguments:

I Do Not Work on MCQA: Our approach to MCQA can apply to all tasks; it is important to question if your format effectively evaluates your intended ability. Educators have long studied the best formats for different tasks, but we are not using these insights to guide our benchmark design. As an example beyond MCQA, in human math assessments, students are often required to "show their work" to verify understanding and diagnose misconceptions (Choy, 2016). However, math datasets like GSM8k (Cobbe et al., 2021) often ignore intermediate computations in their metrics. Similarly, our dataset quality issues are universal; it is always important to ensure datasets are not contaminated or saturated, as well as free from errors and shortcuts. Thus, we advise all researchers—regardless of task or domain—to consult education research to see if they can improve their evaluations' efficacy.

Other modalities, languages, etc. are different: Our critiques of MCQA's rigid goals (§3.1), misalignment with user needs (§3.2), and failure to fully assess knowledge (§3.3), along with our proposed generative task alternatives (§4), are applicable regardless of modality and language. Further, the MCQA dataset quality concerns (§5) we discuss

are still relevant to these domains; for example, in some multi-modal QA datasets, models can answer questions without using the input image (Goyal et al., 2017), showing shortcuts exist (§5.3).

Why not abandon MCQA? Indeed, other formats have grown in popularity, such as prompting LLMs on real user queries and using annotators/-models to judge model responses (Chiang et al., 2024; Lin et al., 2025). These efforts are exciting and directly reflect LLM use cases, but are difficult to scale for every domain we currently use MCQA for, their subjective scoring lacks reproducibility, and these metrics are easy to game (Zheng et al., 2025). In contrast, MCQA and our proposed generative formats include scoring using "pick the best answer", forming a more efficient and objective metric. Thus, we should still aim to advance both of these threads for more reliable evaluations.

This is All Way Too Much Work: We have proposed many directions for future research, but our objective is not to have every MCQA dataset designer engage with each of these efforts. We hope that by pointing out these issues in MCQA, dataset designers will start to consider how using MCQA will affect their datasets' reliability in the long term, and researchers will further study ways to improve MCQA evaluation. Even adopting just one of our proposals could greatly enhance the quality and effectiveness of MCQA datasets. Over time, these small, incremental improvements across the evaluation community will drive meaningful progress.

Generation is too Hard: While generative versions of MCQA are harder to implement, such efforts are warranted to improve the utility of evaluations: generation tasks better test knowledge (§3.3) and mirror LLM use cases (§3.2). We believe the difficulty of implementation should not preclude the adoption or at least exploration of these threads.

We agree MCQA is attractive as it is easy, but this is not the most important property of evaluations; evaluations should measure how the system will behave in deployment (Saxon et al., 2024). LLMs are used to generate text that helps users, so we argue researchers should strive for tasks that measure generation, not just those that are easy to implement. Many fields have also faced difficulty when evaluating their newest systems, like Information Retrieval and Machine Translation, and have thus made progress as a community toward new evaluation datasets (Voorhees, 2001, e.g., TREC), and metrics (Rei et al., 2020, e.g., COMET); we should thus do the same for LLMs with MCQA.

Our proposed shift from validation to generation evaluations is also not totally new. NLP saw a similar trend in summarization, where efforts switched from evaluating if systems could extract sentences within an input text to generating abstractive summaries (Mehta, 2016). Abstractive summarization is harder to evaluate, but we made this change as a community as it better captured the downstream summarization needs of users (Lin and Ng, 2019).

9 Ethical Considerations

Flawed evaluations can mislead both researchers and users; researchers may misinterpret model abilities due to quality issues in datasets, while users may struggle to identify the best models for their needs. This paper outlines several potential solutions to mitigate these risks in MCQA, ensuring more reliable evaluations for researchers and users.

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A Appendix

A.1 Initial Paper Selection Process

To identify relevant papers for our initial reading list, we follow PRISMA (Page et al., 2021), a systematic methodology for paper review. We start by curating 25 keywords related to MCQA evaluation:

- multiplechoice
- multiple-choice
- multiplechoicequestionanswering
- multiple-choice question-answering
- multiple choice
- multiplce choice question answering
- multiple choice evaluation
- multiple choice benchmarks
- multiple choice benchmarking
- multiple choice reasoning
- multiple choice limitations
- multiple choice weaknesses
- multiple choice issues
- multiple choice large language models
- multiple choice llms
- mcqa
- mcqa evaluation
- mcqa benchmarks
- mcqa benchmarking
- mcqa reasoning
- mcqa large language models
- mcqa llms
- mcqa limitations
- mcqa weaknesses
- mcqa issues

We use these keywords to search ArXiv, Semantic Scholar, and ACL Anthology, resulting in 1476 total papers and 1250 unique papers. To help automate the filtering process, we follow Schulhoff et al. (2024) and use `gpt-4o` to classify irrelevant papers. The LLM labels if a paper is "highly relevant", "somewhat relevant", "neutral", "somewhat irrelevant", or "highly irrelevant" by its abstract and title (Prompt A.1). We only keep "highly relevant", "somewhat relevant", or "neutral" papers. We validate the classifier on 200 sampled papers, achieving 92% recall. This filtered 42% of papers.

Post-filtering, we manually screen the remaining 734 papers, excluding 612 studies that only introduce new MCQA benchmarks without providing new findings on model evaluation or focus exclusively on multi-modal MCQA. While we mainly discuss text-only MCQA, many findings are also applicable to multi-modal settings (§8). In total, we used 122 papers to form the initial reading list of this survey, which helped us form our initial arguments. While writing our arguments, we searched for more papers to supplement each of the points we discussed, often from education research.

A.2 Prompts for Examples

On the next page, we provide the prompts used to produce the LLM outputs for all of our figures.

A.3 Additional Related Works

There are several works that expose LLM issues in MCQA (§6), many of which came out around the same time. Due to space constraints, we are unable to include all of them in the main body of the paper. To ensure they are still recognized, we cite these works here. There are several works showing LLM robustness issues in MCQA, studying shuffling option order and formatting perturbations (Zong et al., 2024; Pezeshkpour and Hruschka, 2024; Ranaldi and Zanzotto, 2023; Zheng et al., 2024a; Li and Gao, 2024; Gupta et al., 2025a; Alzahrani et al., 2024; Long et al., 2025; Lyu et al., 2024; Tsvilodub et al., 2024; Khatun and Brown, 2024). Similarly, there are many works showing that LLMs provide unfaithful explanations in MCQA (Agarwal et al., 2024; Kim et al., 2024; Madsen et al., 2024; Lyu et al., 2023; Paul et al., 2024; Turpin et al., 2024).

Prompt A.1: Paper Classifier Prompt

You are a lab assistant, helping with a systematic review on using LLMs to perform MCQA (Multiple Choice Question Answering). Your task is to rate the relevance of a paper to the topic of MCQA, particularly focusing on research related to:
- Format (e.g., limitations of the MCQA format, connection to real-world tasks, assumption of a single best answer).
- Dataset Quality (e.g., saturation, test set leakage, incorrect answers, artifacts and shortcuts).
- Models (e.g., robustness, logical reasoning challenges).

We are not interested in papers related to generating multiple-choice questions.

To clarify:
- Papers focusing explicitly on MCQA methodologies, evaluations, or related challenges are considered highly relevant.
- Papers on closely related topics (e.g., general question answering, NLP datasets, or evaluation methods) may still be relevant if they address concepts transferable to MCQA.
- Papers that solely discuss unrelated NLP tasks, such as translation or summarization, without mentioning MCQA or related issues, are irrelevant.
- Papers focusing on training a model from scratch or using entirely different evaluation paradigms (e.g., open-ended text generation) are also considered irrelevant unless they explicitly tie back to MCQA.

Be aware that a paper might not explicitly spell out "MCQA" but could still use a multiple-choice format or discuss related issues under different terminology. In such cases, use your judgment to evaluate its relevance.

Provide a response in JSON format with two fields:
- 'reasoning': A single sentence justifying your decision.
- 'rating': A string that is one of the following categories:
- 'highly relevant'
- 'somewhat relevant'
- 'neutrally relevant'
- 'somewhat irrelevant'
- 'highly irrelevant'

Prompt A.2: Typical MCQA (Figure 3, top)

Just give the letter of the answer
Question: Heat, light, and sound are all different forms of what?
Choices: (A) fuel (B) energy (C) matter (D) electricity
Answer: (B) energy

Prompt A.3: Constructed Response (Figure 3, middle)

Just give a short form answer
Question: Heat, light, and sound are all different forms of what?
Answer: Energy

Prompt A.4: Justified MCQA (Figure 3, bottom)

Give the letter of the answer and a justification
Question: Heat, light, and sound are all different forms of what?
Choices: (A) fuel (B) energy (C) matter (D) electricity
Answer:
Justification:

Answer: B
Justification: Heat, light, and sound are all different forms of energy. Energy is the ability to do work or produce change. Heat is thermal energy, light is electromagnetic energy, and sound is mechanical energy traveling through a medium. Therefore, the correct answer is B) energy.

Prompt A.5: Checklist Evaluation Prompt (Figure 4)

Here is an MCQ
Question: The number of energy levels for the 55Mn nuclide are:
Choices: (A) 3 (B) 5 (C) 8 (D) 4
Answer: (A)
=====
Go through all of the checklist items and find where the MCQ failed or succeeded:
[pasted checklist from Haladyna and Downing (1989)]

Prompt A.6: Obscure MCQ Prompt (Figure 6, left)

Just give the answer without using the web
Q: In Spongebob, Tony's house has a poster referencing what band?
(A) Queen (B) Gorillaz (C) Kiss
Answer: (A) Queen

Prompt A.7: Adversarial MCQ Prompt (Figure 6, right)

Just give the answer without using the web
Q: How many non-pet characters live in SpongeBob's neighborhood?
(A) 3 (B) 4 (C) 5
Answer: (B) 4

Prompt A.8: LLM Issues — Original (Figure 7, row 1)

Just give the letter of the answer
Question: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Kiss (B) Gorillaz (C) Queen
Answer: A

Prompt A.9: LLM Issues — Robustness (Figure 7, row 2)

Just give the letter of the answer
Question: In Spongebob, Tony's house has a poster referencing what band?
Choices: (P) Kiss (Q) Gorillaz (R) Queen
Answer: R

Prompt A.10: LLM Issues — Bias (Figure 7, row 3, left)

Just give the letter of the answer
Question: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Gorillaz (B) Queen (C) Kiss
Answer: A

Prompt A.11: LLM Issues — Bias (Figure 7, row 3, right)

Just give the letter of the answer
Question: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Queen (B) Gorillaz (C) Kiss
Answer: A

Prompt A.12: LLM Issues — Explanations (Figure 7, row 4)

Just give the letter of the answer
Question: In Spongebob, Tony's house has a poster referencing what band?
Choices: (A) Queen (B) Gorillaz (C) Kiss
Answer: I don't have a record of a character named Tony in SpongeBob SquarePants having a poster referencing a band. However, if you meant Tony Fast Jr. (a minor character from the episode Tutor Sauce), his room has a poster that appears to reference the band Kiss.

Thus, the answer is:

(A) Kiss

I chose this because Kiss is known for their iconic face paint, and the poster in the episode features a character with similar stylized makeup, which aligns with the band's aesthetic.